Hierarchical Index Retrieval-Driven Wireless Network Intent Translation With LLM

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***Abstract*—Intent-Based Networking (IBN) represents an emerging network management concept that is designed to fulfill user service requirements through automation. At its core, IBN is capable of translating user intent into network policies, thereby enabling automated configuration and management. However, the application of IBN has been limited by challenges associated with automation and intelligence. The recent widespread adoption of Large Language Model (LLM) has partially mitigated these issues. Nonetheless, hardware heterogeneity and high dynamic networks remain significant challenges for IBN: *(i)* Devices from different vendors are challenging to manage uniformly; *(ii)* Aligning service demands with rapidly changing network status is difficult. To address these challenges, we propose LIT, a framework of LLM-empowered Intent Translation with manual guidance. LIT incorporates Retrieval-Augmented Generation (RAG) to reference hardware manuals and enhance the generation results of LLMs. To reduce noise from retrieval results, we optimized the general RAG process. Additionally, LIT introduces MoE (Mixture of Experts) to adjust parameter values according to network status by synthesizing results from multiple expert models. Experiments demonstrate that LIT alleviates the challenges faced by IBN, achieving a 57.5% improvement in F1 score compared to the baseline.**

***Index Terms*—Intent-based networking, large language model, intent translation, retrieval-augmented generation.**

I. Introduction

THE advent of sixth-generation communication network technology, coupled with the proliferation of sophisticated access devices and applications, including Virtual Reality (VR), Uncrewed Aerial Vehicles (UAV), and Vehicle-to-Everything (V2X), has placed considerable strain on Network Management Systems (NMS) [1], [2], [3]. In the design of the next-generation network, the development of wireless networks must accommodate a multitude of flexible and diverse service requirements and application scenarios [4]. The NMS is integrating advanced technologies, including artificial intelligence (AI) and machine learning (ML), to facilitate autonomous network management in complex environments [5], [6]. However, due to the differentiated characteristics of users, there is an urgent need within the industry for a convenient tool to achieve an automated external management loop for operators.

The Intent-Based Network (IBN) represents a novel paradigm of network management. Its objective is to realize automated network configuration and operations through the use of abstract intent [7], [8]. As one of the emerging technologies for achieving network autonomous driving, IBN is gradually being standardized by organizations such as the 3 rd Generation Partnership Project (3GPP) [9], the European Telecommunications Standards Institute (ETSI) [10], and the TM Forum [11]. Users need only express their desired network goals or intents, and IBN can automatically translate these intents into specific network configurations and executable policies. Intent translation, as a core component of IBN, plays a crucial role in achieving this goal [12].

The advancement of Natural Language Processing (NLP) technology has enhanced the understanding of abstract intents and the generation of executable strategies, playing a crucial role in intent translation. Traditional intent translation methods typically require extracting entities from the intent and matching them with policy templates [13]. However, models like RNNs and LSTM networks have limited capabilities and rely heavily on extensive supervised data [14]. While Pre-trained Language Models (PLMs) such as BERT [15] perform well in entity extraction with fine-tuning on smaller datasets, most intents do not contain all the entities required for policy generation. To achieve fine-grained intent understanding and strategy generation, language models (LLMs) with robust comprehension and generation abilities are particularly crucial.

In recent years, LLMs such as GPT [16] and GLM [17], with their powerful self-supervised learning capabilities and deep neural network architectures, have made significant progress in understanding and generating complex texts. This ability to capture syntactic, semantic, and contextual relationships from

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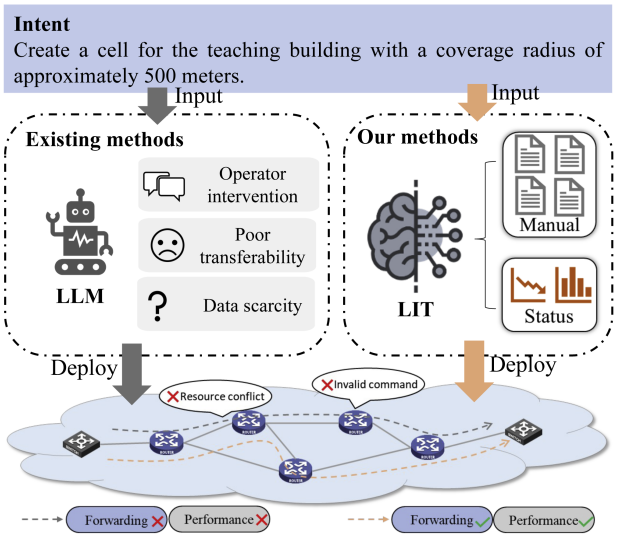


Fig. 1. Comparison of existing methods and LIT. Existing methods achieve intent translation end-to-end through LLMs, with issues in the gray boxes. These issues directly lead to network forwarding failures or resource allocation conflicts.

unlabeled text enables the large models to efficiently parse unstructured intents and dynamically infer implicit constraints in network management scenarios. This has enabled LLMs to demonstrate unique adaptability in fields such as network management, providing promising solutions for intent translation [18], [19].

However, this promising solution merely addresses some semantic-level issues. A truly crucial problem remains that limits the development of IBN: *how to adapt existing intent translation schemes to the intricate environment of hardware heterogeneity and dynamic networks.* This problem manifests in three key challenges:

- *C1 The lack of intent labeled data based on heterogeneous hardware:* Intent execution typically requires multiple policies that are grammatically correct and executed in the correct sequence, which makes obtaining labeled data extremely challenging. Under complex conditions, policies written by experienced operators may contain errors.
- *C2 The legality of generating Parameter Values (PVs):* Manuals specify the types and ranges of PVs, and some parameters depend on the values of other parameters. These complex constraints create significant difficulties in generating PVs.
- *C3 Alignment of generated policies with network status:* For example, the generation of metric PVs such as transmission power and bandwidth must take into account user demands and the network status. The proximity of wireless networks to business users results in a more diverse range of services, necessitating a correspondingly more varied set of requirements.

The comparison between existing methods and LIT is illustrated in Fig. 1. These challenges highlight the need for a more robust and adaptive intent translation framework, which LIT aims to address.

The intent translation framework, LIT, is introduced to address the aforementioned challenges. LIT consists of two stages: Policy Sequence Generation (PSG) and Parameter Value Generation (PVG). To tackle ***C1*** and ***C2***, LIT incorporates manual learning and Retrieval-Augmented Generation (RAG) to assist the LLMs in completing the intent translation task. Based on the characteristics of the hardware manual, two new issues have arisen: (1) how to enable LLMs to understand complex parameter dependencies, and (2) too much relevant text retrieved, leading to excess noise, which negatively affects LLM comprehension. Therefore, we optimized the general RAG process specifically for the intent translation task. This includes constructing a hierarchical index for manuals to facilitate the retrieval of relevant sentence chunks and improve LLM understanding of parameter dependencies. Additionally, we quickly compress the retrieved text to reduce noisy information. To address ***C3***, we introduce an MoE (Mixture of Experts) based on LLMs, which understands user business requirements through text-based interaction. This enables more effective coordination of the collaboration of expert models. This approach benefits the LLMs in the unified scheduling of experts, solving complex optimization problems, and generating more precise PVs. In comparison to training and optimizing models for each user's requirements individually, MoE markedly diminishes the computational resources utilized for model training and deployment.

In conclusion, the main contributions are as follows:

- We propose a novel intent translation framework, LIT, for wireless networks. LIT leverages the understanding and generation capabilities of LLMs to efficiently translate the abstract intents into specific sets of policies. LIT integrates manual knowledge and expert experience to achieve fine-grained alignment between service requirements and network status.
- We propose an effective generation method guided by reference manuals. This method enhances the generalization ability of LLMs in network domain tasks by learning from hardware manuals, enabling the generation of logically and grammatically correct Policy Sequence (PS). RAG helps LLMs understand complex parameter constraints, thereby generating more precise PVs.
- We optimized the process of RAG by proposing a retrieval method adapted to manuals. This method is based on automatically constructing an index tree from manuals, compressing retrieval results, and effectively assisting the LLM in understanding complex parameter dependencies.
- We propose a parameter value generation method based on MoE, utilizing LLMs to precisely model user requirements and considering decisions from multiple expert models to derive the final result. This method demonstrates its effectiveness by generating metric-level PVs affecting user service, thus achieving precise matching of user requirements with network resources.

II. Related Work

A. Intent-Based Network

IBN represents a novel network paradigm, distinguished by its capacity to autonomously convert, verify, deploy, configure,

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and optimize itself in a manner that aligns with the desired network state, as defined by user intent [20]. Furthermore, it is equipped with the capacity to autonomously identify and rectify irregularities, thereby guaranteeing the dependability of the network [6], [21].

Lumi [22] translates operator intents into policies through the abstract language Nile [23]. However, in complex scenarios involving temporal constraints or multi-device coordination, this method struggles to accurately capture contextual semantics. Additionally, the template-driven compilation mechanism limits its adaptability to cross-vendor device configurations, making it even more challenging to meet user demands under heterogeneous conditions. To address the issue of device heterogeneity across different providers, [24] and [25] both proposed a configuration generation framework based on hardware manual learning. By learning from the hardware manuals, the generated configurations align with the syntactical requirements of various devices, thereby enhancing robustness. However, intent translation often requires consideration beyond the semantic level, as the dynamic changes in the resources of downstream networks are also critically important. Chroma [26] learns the context after the exploitation phase and recommends configuration changes using the contexts for similar locations. Chroma effectively optimizes performance based on past changes but is overly focused on network self-optimization, lacking user interaction. In recent years, with the widespread application of LLMs in network tasks, studies such as CoSynth [27] have attempted to generate configuration commands by invoking GPT-4 [27] and correct them through multi-layer validators. Although such methods have made progress in syntax error correction, they inherently decouple domain expertise from network dynamics: generic models misinterpret topology-specific constraints, while static validators fail to integrate real-time operational states, resulting in policies that are misaligned with actual network conditions.

Wireless networks are closer to commercial users and carry a vast amount of network services and devices, which presents greater challenges for network management, particularly in intent translation [28]. LIT is capable of effectively understanding user intent and subsequently generating network policies. Furthermore, it addresses the issue of device heterogeneity through the RAG, thereby producing high-quality network policies while avoiding conflicts in network resources.

B. Large Language Model

LLMs represent a substantial technological advancement in the domain of natural language processing, exhibiting noteworthy advancement in recent years. The core concept of these models entails training with extensive datasets and deep learning algorithms to develop models with robust capabilities in natural language understanding and generation [16], [19]. They have demonstrated excellent performance across multiple application domains, including text generation, machine translation, dialogue systems, content recommendation, and more. As the scale of the models increases, their performance has seen significant enhancements. In industries such as search engines, autonomous driving, intelligent customer service, and personalized recommendations, LLMs have become indispensable technological support [29].

The integration of LLMs into communication networks has catalyzed a paradigm shift toward intelligent NMS [30]. LLMs can be applied to tasks such as network configuration, fault diagnosis, and resource optimization, significantly improving the efficiency and accuracy of network operations [31]. Furthermore, the ability of LLMs to process multimodal data and interact with domain-specific knowledge bases enhances the adaptability of network management frameworks, fostering autonomous decision-making and proactive optimization [32].

The black-box nature of LLM makes their reasoning processes difficult to interpret [33], [34]. RAG has been proposed as an effective solution to this issue in recent years [35], [36]. RAG integrates information retrieval and generation models to enhance the precision and comprehensiveness of knowledge in generation tasks [37], [38]. The workflow of RAG generally includes two main steps: retrieval and generation. (1) Retrieval: the system receives an input query and employs a retriever to extract the most pertinent information from a knowledge base. This information may encompass text fragments, documents, and other forms of data. (2) Generation: the generation model synthesizes the retrieved content with the original query to generate the final output [35].

C. Mixture of Experts

The MoE framework represents a technique rooted in deep learning, aimed at managing and refining experts. This is achieved by distributing tasks to various sub-models, each tailored to specialize in particular domains [39]. The advantage of MoE lies in its capacity to scale models to accommodate complexity and disparate data types or tasks. This contrasts with conventional single-model architectures, which are constrained by computational and memory limitations [40].

MoE has previously been widely applied in computer vision and natural language processing [41], [42]. It operates with a gating mechanism that identifies and activates the most relevant experts from a pool, guided by the input data, to collectively arrive at decisions. This allows either all experts to participate or only a specific subset. The methodology is particularly advantageous because it enables the reuse of the expert pool across different tasks. Only the gating mechanisms require training for each new task, offering a balance of versatility and efficiency [43]. With the vast range of network services available, creating and implementing separate models for each service or user requirement is not feasible. Such a process not only consumes excessive resources but also hampers model adaptability [44]. Consequently, this paper introduces an MoE approach leveraging LLMs to identify suitable expert models, ensuring efficient decision-making based on the LLM's comprehension of service demands.

III. Problem Description

As the natural language vehicle for expressing user requirements in IBN, the forms of expression of intents are diverse.

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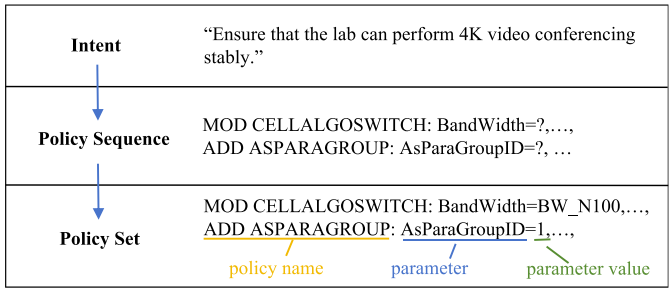


Fig. 2. The example of intent, PS, and policy set. Intent refers to the user's natural language input. Policy set is a set of ordered policies that can be deployed by network controller to achieve the intent. PS represents a policy set that does not include PVs.

The goal of IBN is to translate the abstract natural language intents of users into executable network configurations and deployments, thereby increasing the user's flexibility in managing networks [12], [45]. Due to the complexity of intents, the realization of a single intent often requires multiple policies, each containing several parameters. Policy sets often encompass dozens or even over a hundred parameters, with PVs being subject to various constraints. A two-stage intent translation process is shown in Fig. 2. The policy set is the output of intent translation, composed of a group of executable policies. Each policy consists of three parts: a policy name, several parameter names, and PVs. The PS is an intermediate state between intent and policy set, differing from the policy set in that it does not contain PVs.

Device manuals specify various properties of policies, including descriptions of the functions that policies implement, the descriptions of parameters, the range and types of PVs, and other content. PVs may be constrained by three aspects: (1) the value range specified in the manual, (2) dependencies on other parameters, and (3) the complex network status. The semantic-level parameters often have intricate dependencies, for example, when parameter A takes the value of 1000–2000, parameter B has a range of 500-1000. Hence, parameters can be classified into two categories based on whether they are influenced by the downstream network: (1) semantic-level parameters, including *LocalCellId* and *LocalCellName*, and (2) metric-level parameters, such as *BandWidth* and *TransPower*.

To achieve rational resource allocation and generate metric-level PVs, it is necessary to consider the status of downstream wireless networks, such as bandwidth. The network nodes consist of user terminals, wireless base stations (primarily macro base stations), and switches. The notation of the variables can be found in Table I. The downstream network topology is described as: $G = \{N, L\}$, with N nodes and L links. Each link l ($l \in L$) is assigned a specific bandwidth resource D_l. The network aims to satisfy all user services $\mathcal{V}$ by allocating link bandwidth resources D_v to the user service v ($v \in \mathcal{V}$). Each user service v includes the source node s_v and the destination node b_v. L_{s_v,b_v} denotes the set of all links along the path between the two nodes. Various Quality of Service (QoS) indicator values

TABLE I
NOTIONS OF VARIABLES

Symbol	Description
$G(N, L)$	A downstream wireless network consists of N nodes and L links.
$\mathcal{V}$	All services in the network.
s_v	The source node of service v.
b_v	The destination node of service v.
L_{s_v,b_v}	The set of all links along the path between the s_v and b_v.
D_l	The bandwidth resource of link l.
D_v	Bandwidth resources allocated to v.
D_a	A available bandwidth value settings.
F	Various QoS indicators contained in v.
f_k	The individual QoS indicator.
$\mathcal{F}_k(\cdot)$	The calculation method of f_k.
$f_{k,\max}$	The maximum QoS indicator f_k that the current network can provide to the user.
$f_{k,\min}$	The minimum acceptable QoS for the service.
Y_v	Whether the currently allocated bandwidth meets all QoS indicator requirements of v.
R_{v,f_k}	The benefit of f_k in each v.
$H(\cdot)$	Utility function of standardized evaluation framework.
$\mathcal{S}_v$	The state of the network for MDP.
$\mathcal{A}$	The action space of MDP.
r_v	The reward of MDP.
a_v	The action can be taken for service v.
$\hat{a}$	The ultimately chosen action for the service.
W	The weight matrix to the selected k expert models.
$Q(\cdot)$	Q value vector based on the input network state.
$\hat{Q}(\cdot)$	The ultimate weighted Q value vector.
$\mathbf{z}^{\text{doc}}$	The manuals documents.
$\mathbf{z}^{\text{que}}$	The query prompt set.
α	The compression rate.
ϱ_m	A metric to evaluate the importance of document $\mathbf{z}_m^{doc}$.
$\ell_{\mathbf{z}_i}$	A metric based on contrastive perplexity to measure the relevance of tokens in the document $\mathbf{z}_i$ to the question.

included in the service v are denoted by $F = \{f_1, f_2, ..., f_k\}$. There is a corresponding calculation formula for each QoS indicator value: $\{\mathcal{F}_1(\cdot), \mathcal{F}_2(\cdot), \ldots, \mathcal{F}_k(\cdot)\}$. The $\mathcal{F}_k(\cdot)$ can be calculated based on [46], [47]. The network generates rewards based on the Service Level Agreement Satisfaction Ratio (SSR) and QoS benefits gained after delivering services to multiple users. To calculate SSR, a binary variable $Y_v \in \{0, 1\}$ is defined to indicate whether the allocated bandwidth satisfies all QoS requirements of the service. Consequently, the overall SSR of the network is defined as:

$$SSR = \frac{\sum_{v \in \mathcal{V}} Y_v}{|\mathcal{V}|} \tag{1}$$

The distribution of bandwidth resources may result in some degree of link resource wastage, which can degrade network performance by increasing latency or packet loss. This inefficiency tends to grow progressively as the demand on link bandwidth resources intensifies [48]. Taking the above factors into account, we establish the method by which the network evaluates the benefit of each individual QoS indicator f_k for each service

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requirement v:

$$R_{v,f_k} = \eta_1 H\left(\mathcal{F}_k\left(D_v, \mathcal{S}_v\right)\right) - \eta_2 \sum_{l \in L_{s_v,b_v}} \frac{D_v}{D_l} \tag{2}$$

where η_1 signifies the user payment obtainable by satisfying QoS indicators. $\mathcal{S}_v$ denotes the resource status of the links traversed by the user to reach the server. η_2 is the cost coefficient, indicating the link loss incurred by the network when allocating bandwidth. The utility function, represented as $H(\cdot)$, computes the present QoS metric value based on the allocated bandwidth and normalizes this metric to assess the extent to which the service's QoS requirements are satisfied. This normalized value is subsequently utilized to help determine the service fee that the respective user need to pay. The maximum QoS indicator f_k that the current network can provide to the user is denoted as $f_{k,\max}$, whereas the minimum acceptable QoS for the service is denoted as $f_{k,\min}$. Therefore, the utility function $H(\cdot)$ can be calculated as follows:

$$H\left(\mathcal{F}_k\left(D_v, \mathcal{S}_v\right)\right) = \frac{\mathcal{F}_k\left(D_v, \mathcal{S}_v\right) - f_{k,min}}{f_{k,max} - f_{k,min}} \tag{3}$$

In conclusion, the total revenue of network is defined by the following optimization function:

$$\max_{\boldsymbol{D_v}} \quad \lambda_1 SSR \cdot \tau + \lambda_2 \sum_{v \in \mathcal{V}} \sum_{f_k \in F} R_{v,f_k} \tag{4a}$$

$$\text{s.t.} \quad D_v \le D_l, \forall l \in L_{s_v,b_v} \tag{4b}$$

$$D_v \in D_a \tag{4c}$$

$$\mathcal{F}_k(D_v, \mathcal{S}_v) \ge f_{k,min} \tag{4d}$$

where λ_1 and λ_2 are coefficients denoting the relative importance of SSR rewards and QoS benefits, respectively. τ represents the additional unit service fee accrued by the network from the user, contingent upon the satisfaction of the SSR. D_a={d_1,d_2,...,d_a} denotes the set of available bandwidth values. Equation (4b) aims to ensure that the assigned bandwidth does not surpass the remaining available resources on each link. Equation (4c) regulates the range of optional bandwidth parameters. Equation (4d) guarantees that the allocated bandwidth satisfies the specific QoS requirements of the service. To prevent resource conflicts and maximize network benefits, appropriate bandwidth allocation is necessary.

IV. Method

A. Overview

The main process of LIT can be divided into two phases: PSG and PVG, as illustrated in Fig. 3. The PSG phase involves offline fine-tuning of an LLM using annotated data and operation manuals to create the PSG, followed by online translation of user intents into structured PS. The PVG phase operates through dual pathways: offline preparation constructs a Hierarchical Knowledge Base (HKB) by segmenting manuals into sentence chunks via rule-based processing, while online execution generates semantic-level PVs through RAG from the HKB and resolves metric-level PVs using a MoE model with domain-specific rules.

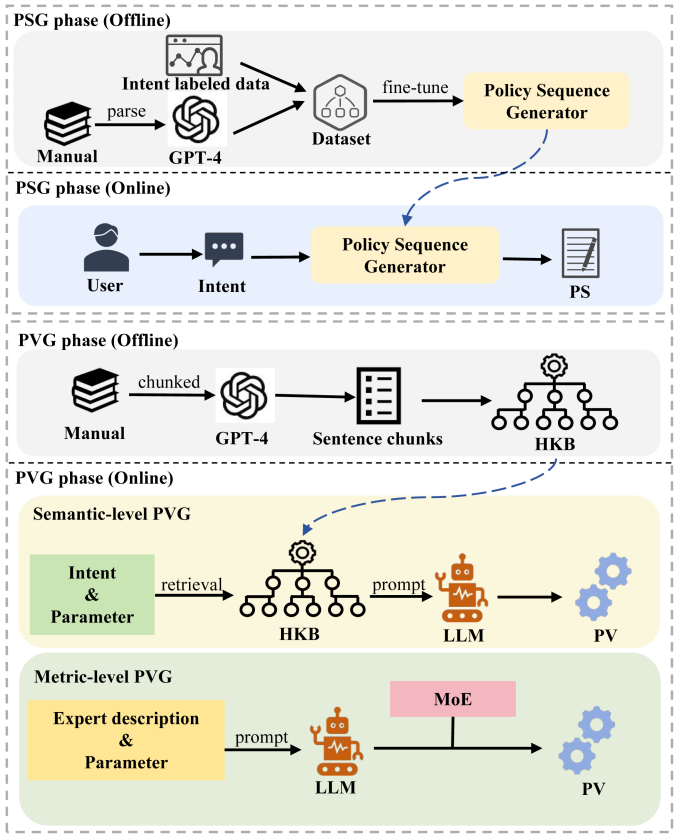


Fig. 3. The main process of LIT. After the user propose an intent, the PSG outputs the PS. Depending on the types of the parameters, PVG generate the PVs.

PSG Phase (offline)
Your task is to organize the provided manual files into a dataset. The input of the dataset is a question inquiring about the function of a specific policy, and the output is a description of the policy's function, which needs to be retrieved from the manual.
Now, think step by step and output the result.

PVG Phase (offline)
Now, break the manual files into sentence chunks, with their indices assigned based on the section titles. For example, if a sentence chunk belongs to Chapter 1, Section 2, Subsection 3, its index should be <123>.
The output should follow this format:
{<index>, <content>}

Fig. 4. The prompt for GPT-4 during the offline phases of PSG and PVG.

This phased architecture ensures modular controllability across the pipeline. Fig. 4 illustrates the prompt for GPT-4 during the offline phases of PSG and PVG.

The capabilities of LIT to understand intent, comprehend policy generation, and generate policy have been significantly enhanced by the outstanding performance of LLM. LIT is capable of accurately comprehending user intents and uncovering the fine-grained resource requirements embedded within those

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TABLE II
DETAILS OF THE FINE-TUNING DATASETS

Dataset	Input	Output
Sentence pair of policy name and description	What is the role of ADD CELL?	This command is used to create a new cell.
Sentence pair of parameter and description	What does PhyCellId mean?	This parameter represents the physical cell identity...
Sentence pair of policy and parameter	What are the parameters contained in the instruction ADD ASPARAGROUP?	The ADD ASPARAGROUP command contains the following parameters: AsParaGroupID,...
Sentence pair of intent and PS	Ensure that the lab can perform 4K video conferencing stably.	MOD CELLALGOSWITCH: LocalCellId=1, BandWidth=...

intents. Additionally, the policy generation process incorporates the network status as a primary factor, enabling dynamic adjustments to resource allocation in alignment with service objectives and the prevailing network status.

B. PSG Phase

The content of intents is often abstract and ambiguous, yet a single intent typically requires multiple policies to achieve the desired goal [49]. For example, to set up a base station suitable for a school area, operations must be carried out in a specific sequence. This includes creating sectors, setting up the base station, configuring algorithms and activating the base station. In order for the PSG to work correctly, it must understand the goal that the intent is trying to achieve. It must also be able to match the objective with relevant policies and their parameters.

Given the scarcity and complexity of annotated data for intent translation tasks, we use GPT-4 to organize manuals and extract parameter and policy-related corpus data. This corpus, combined with intent-labeled data, is used to fine-tune the PSG. A comprehensive overview of the relevant datasets is presented in Table II. Data extracted from the manuals can be used as an auxiliary task to improve the performance of the intent translation task. These datasets provide the necessary annotated data for fine-tuning the LLM, enabling it to understand manual instructions and generate executable policies. We trained the Policy Sequence Generator using an end-to-end full-parameter fine-tuning approach.

During the online phase, the PSG generates PS based on user input in natural language intents. By learning from the manuals, the LIT's logical reasoning capability in generating PS is enhanced, thereby reducing the dependency on large amounts of labeled data.

C. Semantic-Level PVG Based on RAG

1) Workflow: The semantic-level PVG is constrained by the manual, including the types of values and their respective ranges. Using the manual as training data for the models significantly increases the computational cost and lengthens the training cycle. In order to address this issue, LIT introduces RAG, using manual searches to adapt to different heterogeneous hardware devices without the necessity of long time training. However, the most manuals present two challenges: (1) The manuals contain numerous similar function names and analogous descriptions of their functionalities, which leads to biases in the retrieval results. (2) The large volume of retrieved sentence chunks contains excessive noise, hindering the LLM's ability to extract information and generate PVs. LIT has optimized the RAG process by implementing indexed retrieval, which facilitates the acquisition of relevant texts. These texts are then compressed to minimize the presence of extraneous data. The workflow of the semantic level PVG is illustrated in Fig. 5. The retrieval algorithm identifies the top-k relevant sentence chunks from the HKB based on the query. These retrieved sentence chunks are then concatenated to form the original query, which is then compressed and fed into the LLM to generate the PVs.

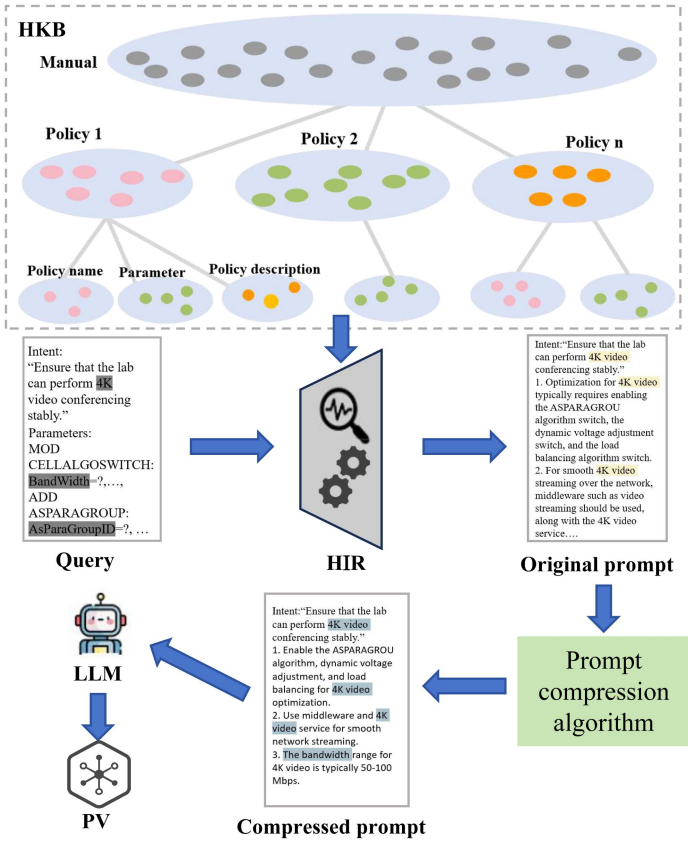


Fig. 5. The workflow of semantic-level PVG.

2) Hierarchical Index Retrieval Algorithm Based on Manual: To retrieve relevant documents using RAG, the Hierarchical Index Retrieval (HIR) algorithm contains two parts: hierarchical manual indexing and retriever training. The former involves the generating specific identifiers associated with each manual documents. The latter involves the construction of a text-to-text model and its subsequent fine-tuning using an available set of labeled data consisting of queries, relevant manual documents and corresponding identifiers (For example, if a identifier is '123', 1 represents the manual, 2 represents Policy 2, and 3

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Algorithm 1: Hierarchical Manual Indexing and Retriever Training.

Input: Manual documents $\mathbf{z}^{\text{doc}}$, pre-trained weights of text-to-text model ω, and query prompt set $\mathbf{z}^{\text{que}}$
Output: Corresponding identifier strings $\mathcal{I}$ and trained weights ω^*

Function `GetIdentifiers(`$\mathbf{z}^{doc}$, n`)`**:**
  $C_{1:n} \leftarrow$ `Cluster`$(\mathbf{z}^{\text{doc}}, n)$;
  Initialize identifier list $\mathcal{I}$;
  for $index = 0$ **to** $n-1$ **do**
    $\mathcal{I}_{\text{now}} \leftarrow [index] * |C_{index+1}|$;
    if $C_{index+1}$ *can divided* **then**
      $u =$ `Classification`$(C_{index+1})$;
      $\mathcal{I}_{\text{rest}} \leftarrow$ `GetIdentifiers`$(C_{index+1}, u)$;
    else
      $\mathcal{I}_{\text{rest}} \leftarrow [0, ..., |C_{index+1}| - 1]$;
    $\mathcal{I}_{\text{cluster}} \leftarrow \mathcal{I}_{\text{now}} + \mathcal{I}_{\text{rest}}$;
    $\mathcal{I}$.append(J_{cluster});
  return J;
Initialize the model with pre-trained weights ω;
$\mathcal{I} \leftarrow$ `GetIdentifiers` $(\mathbf{z}^{\text{doc}})$;
Construct an available set of labeled training data, $(\mathbf{z}^{\text{que}}, \mathbf{z}^{\text{doc}}, \mathcal{I})$;
Set optimizer with learning rate and loss function;
for *epoch* $= 1$ **to** E **do**
  for *Each batch in training data* **do**
    Compute loss $\mathcal{L} = \text{CrossEntropy}(I_{\text{pred}}, I_{\text{true}})$;
    Compute gradients $\nabla_\omega \mathcal{L}$;
    Update weights: $\omega \leftarrow \omega - \eta \nabla_\omega \mathcal{L}$;
return identifier strings $\mathcal{I}$ and the trained weights ω^*;

represents Parameter 3.) [50]. The retriever is capable of generating a sequence of identifiers associated with a specific query. Ultimately, beam search [51] is employed to evaluate the top-k identifiers and identify the manual documents that correspond to the identifiers, which are then presented as the retrieval results. The process of hierarchical manual indexing and retriever training is elaborated in Algorithm 1. $\mathbf{z}^{\text{doc}}$ represents the manuals documents and $\mathbf{z}^{\text{que}}$ represents the query prompt set.

The implementation of hierarchical manual indexing can markedly diminish the occurrence of superfluous information during the retrieval process, thereby optimizing both the precision and recall of manual document retrieval. The generation of identifiers for hierarchical manual indexing necessitates the utilization of a hierarchical clustering process based on manual document embeddings, which ultimately yields an n-ary index tree. The manual categorizes policy documents based on functional differences, and based on this categorization, different policy documents are divided into n clusters. Each policy document is assigned an identifier containing its cluster number, ranging from 0 to $n-1$. Within each policy document, various descriptive documents can be further subdivided into smaller clusters, such as policy name, parameters, and policy description. Each sub-cluster is divided into u classes based on the descriptive documents and is assigned a number between 0 and $u-1$. If the documents within the sub-cluster can be further divided, the sub-clustering process is iterated. Within each cluster, each element is assigned a number ranging from 0 to the total number of documents in the cluster minus 1, which is then appended to the existing identifier. This process ultimately completes the classification of the manual documents and assigns a specific identifier to each document, forming the hierarchical manual indexing.

3) Prompt Compression Algorithm: After completing retrieval, a prompt is created based on the retrieval results and the target parameter, and then the LLM generates PVs based on the prompt. However, overly long prompts can introduce excessive noise, hindering the LLM's ability to extract useful information effectively. To address this issue, we propose a prompt compression algorithm tailored for LLMs. This algorithm significantly reduces irrelevant textual information in the documents while preserving content related to PVs, thereby greatly improving the accuracy of PVs generation.

The prompt is represented by $\mathbf{z} = (\mathbf{z}^{\text{ins}}, \mathbf{z}_1^{\text{doc}}, \ldots, \mathbf{z}_M^{\text{doc}}, \mathbf{z}^{\text{que}})$, including the instruction $\mathbf{z}^{\text{ins}}$, M documents $\mathbf{z}_i^{\text{doc}}$, and the question $\mathbf{z}^{\text{que}}$. The goal of prompt compression can be expressed as:

$$\min_{\tilde{\mathbf{z}}} D_\phi \left(\mathbf{X}, \tilde{\mathbf{X}} \right) + \beta \|\tilde{\mathbf{z}}\|_0, \tag{5}$$

where $\tilde{\mathbf{z}}$ denotes the condensed prompt, which is a token-level subsequence derived from $\mathbf{z}$. The symbols $\mathbf{X}$ and $\tilde{\mathbf{X}}$ denote the outputs generated by the LLM from $\mathbf{z}$ and $\tilde{\mathbf{z}}$, respectively. The function D_ϕ is utilized to evaluate the distance metric, such as Kullback-Leibler divergence. The parameter β is used to adjust the balance of the compression rate. The compression rate is defined by $\alpha = \frac{length(\tilde{\mathbf{z}})}{length(\mathbf{z})}$, where α is bounded between 0 and 1, and the compression ratio is represented as $\frac{1}{\alpha}$.

The prompt compression algorithm consists of two stages by using a small LLM to assess the perplexity for each token in the prompt: coarse-grained compression and fine-grained compression. In the coarse-grained phase, a metric is calculated to evaluate the importance of each document, retaining only those documents with a higher ranking as an intermediate compression result. Additionally, a constraint statement describing $\mathbf{z}^{\text{res}}$ is introduced to strengthen the connection between the question and the documents. The specific calculation of the metric is as follows:

$$\varrho_m = -\frac{1}{N_c} \sum_{i}^{N_c} \log p \left(\mathbf{z}_i^{\text{que, res}} \mid \mathbf{z}_m^{\text{doc}} \right), m \in \{1, 2, \ldots, M\} \tag{6}$$

where $p(\mathbf{z}^{\text{que,res}} \mid \mathbf{z}_m^{\text{doc}})$ describes the conditional probabilities of the question $\mathbf{z}^{\text{que,res}}$ given various contextual conditions $\mathbf{z}_m^{\text{doc}}$. This metric is utilized to signify the direct relationship between each document and the question. Here, $\mathbf{z}_i^{\text{que,res}}$ represents the i-th token within the combined sequence of $\mathbf{z}^{\text{que}}$ and $\mathbf{z}^{\text{res}}$, with N_c denoting the total number of tokens. Following this, documents are ordered based on these metric values, and the top M' are chosen as the outcome of the coarse-grained compression

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process:

$$\left(\mathbf{z}^{\text{ins}}, \mathbf{z}_1^{\text{doc}}, \ldots, \mathbf{z}_{M'}^{\text{doc}}, \mathbf{z}^{\text{que}}\right) \xrightarrow{\varrho_m} \left(\mathbf{z}^{\text{ins}}, \mathbf{z}_{\varrho_1}^{\text{doc}}, \ldots, \mathbf{z}_{\varrho_{M'}}^{\text{doc}}, \mathbf{z}^{\text{que}}\right) \quad (7)$$

Fine-grained compression predominantly employs contrastive perplexity, which assesses the distributional change induced by the condition of the question, to evaluate how relevant the tokens in the document are to the question. And the perplexity is used to quantify the capability of a language model to understand and generate a piece of text. It is derived from computing the exponential result of the cross-entropy loss for a given text. The importance metric $\ell_{\mathbf{z}_i}$, which relies on contrastive perplexity for each token $\mathbf{z}_i$ within the document, can be expressed as follows:

$$\ell_{\mathbf{z}_i} = \text{perplexity}\left(\mathbf{z}_i \mid \mathbf{z}_{<i}\right) - \text{perplexity}\left(\mathbf{z}_i \mid \mathbf{z}^{\text{que}}, \mathbf{z}_{<i}\right). \quad (8)$$

Given the variability in the density of key information across different documents, it is reasonable to conclude that documents more pertinent to the question at hand should be assigned a higher compression rate. Hence, a connection between coarse-grained and fine-grained compressions can be established, utilizing the metric value ϱ_m from the former to direct the allocation of compression rates in the latter. For documents that have undergone coarse-grained compression, the compression rate α_m^{doc} is adaptively assigned to each document $\mathbf{z}_m^{\text{doc}}$ according to the ranking index $I(\varrho_m)$ (such as 0 or 1) produced during the coarse-grained compression. Consequently, the compression rate assigned to each token z_i can be described as:

$$\alpha_i = \alpha_m^{\text{doc}}, \quad \forall \mathbf{z}_i \in \mathbf{z}_m^{\text{doc}} \quad (9a)$$

$$\alpha_m^{\text{doc}} = \max\left(\min\left(\left(1 - \frac{2I(\varrho_m)}{M'}\right)\delta_\alpha + \alpha_m^{\text{doc}}, 1\right), 0\right) \quad (9b)$$

where i and m refer to the indices of the token and document respectively, M' signifies the total number of documents, and δ_α serves as a hyper-parameter regulating the overall compression rate for dynamic assignment.

After obtaining the compression rate for document $\mathbf{z}_m^{\text{doc}}$ and the distribution of importance metric $\ell_{\mathbf{z}_i}$ for each token within the document, the compression rate threshold $\mathcal{T}_m$ for $\mathbf{z}_m^{\text{doc}}$ can be dynamically calculated. Finally, tokens in the document $\mathbf{z}_m^{\text{doc}}$ with $\ell_{\mathbf{z}_i,m}$ greater than $\mathcal{T}_m$ will be retained in the compressed prompt:

$$\tilde{\mathbf{z}}_m^{\text{doc}} = \{\mathbf{z}_{i,m} \mid \ell_{\mathbf{z}_i,m} > \mathcal{T}_m\} \quad (10)$$

where $\mathbf{z}_{i,m}$ denotes the i-th token in document $\mathbf{z}_m^{\text{doc}}$.

D. Metric-Level PVG Based on MoE

IBN oversees the management of downstream networks, translating intents into network policies that include the allocation of specific network resources. The generation of network policies is based on intents that encapsulate service requirements, with the fluctuating state of downstream networks also exerting an influence. It is therefore essential that metric-level PVG aligns upstream service demands with the status of downstream networks in order to maximize total revenue while avoiding resource conflicts. Furthermore, given that networks typically encompass a large number of users, training a separate model for each user or service is highly resource-intensive and inefficient, as these models are difficult to adapt or generalize to new services.

Algorithm 2: The Expert Process of ECOS Solver.

Input: The user service v, the source node s_v and the destination node b_v of the service request
Output: The bandwidth resources allocated to each user D_v

```
Initialize MILP model with decision variables D_v;
Define the objective function R_{v,f_k} via Eq. (2);
Add constraints D_v ≤ D_l, ∀l ∈ L_{s_v,b_v};
Add constraints F_k(D_v, S_v) ≥ f_{k,min} ;
Add variable bounds D_v ∈ D_a = {d_1, d_2, ..., d_a} ;
for v = 1, ..., V do
    Calculate the remaining bandwidth D_s of all links
     between s_v and b_v ;
    Set the linear solver ECOS ;
    if solver finds a feasible solution then
        Extract optimal solution D_v^* ;
        Allocate the bandwidth D_v^* to the users ;
    else
        Report that no feasible solution is found;
```

LIT employs LLM as a gating mechanism to analyze intents and generate a weight matrix based on the optimization objectives of the expert models. Thereafter, each expert model determines its own resource allocation results. Ultimately, the final resource allocation values are determined by combining the weight matrix with the results from each expert model. The workflow of the metric-level PVG based on MoE is illustrated in Fig. 6. Two approaches are employed for training expert models: (1) conversion of the optimization problem into a Mixed Integer Linear Programming (MILP) problem, with the linear solver ECOS [52] utilized to construct the expert model for solving the MILP problem; and (2) construction of a Markov Decision Process (MDP), with Deep Reinforcement Learning (DRL) algorithms employed for training the expert model.

The objective of each expert model is to identify the optimal bandwidth configuration that maximizes the benefit of a specific QoS indicator. For the construction of expert models using mathematical solvers, a MILP model is constructed based on (2) as the objective function and constraints (4b), (4c), and (4d). The ECOS solver is used to select the optimal bandwidth value from the available bandwidth options to maximize the objective function. The expert process of ECOS solver is detailed in Algorithm 2. For training expert models using DRL, an MDP is constructed in which the network makes bandwidth allocation decisions based on the current network state. As the next user request arrives, the resulting network state depends only on the previous state and action, conforming to the Markov property. In this study, the Deep Q-Network (DQN) [53] is utilized as the expert model.

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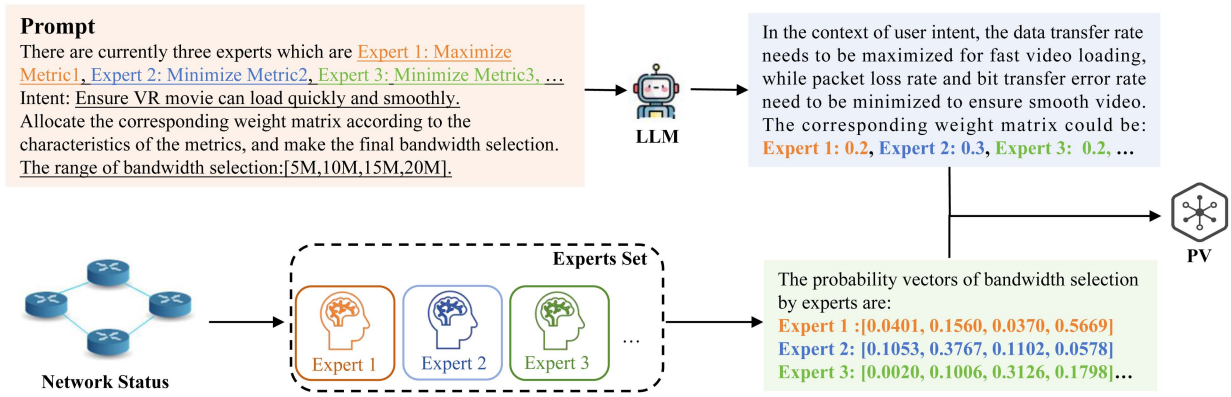


Fig. 6. The workflow of the metric-level PVG based on MoE. The Prompt consists of the expert's optimization directions, intent, and required operations. Based on the received Prompt, the LLM outputs an expert's weight matrix, which is then multiplied with the expert's results to generate the PVs.

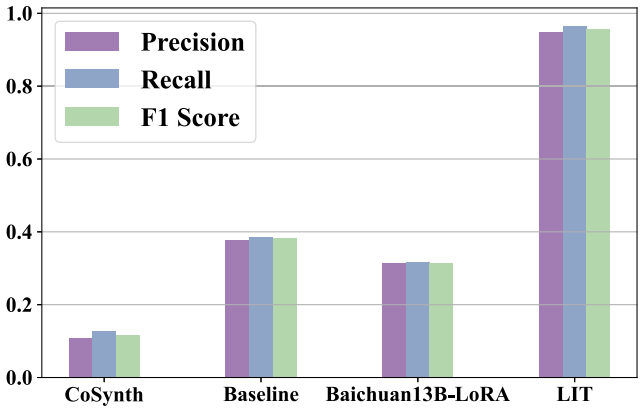


Fig. 7. The comparison of the Precision, Recall, and F1 Score results for intent translation between CoSynth, ChatGLM-6B (baseline), Baichuan13B-LoRA, and LIT on the LIT dataset.

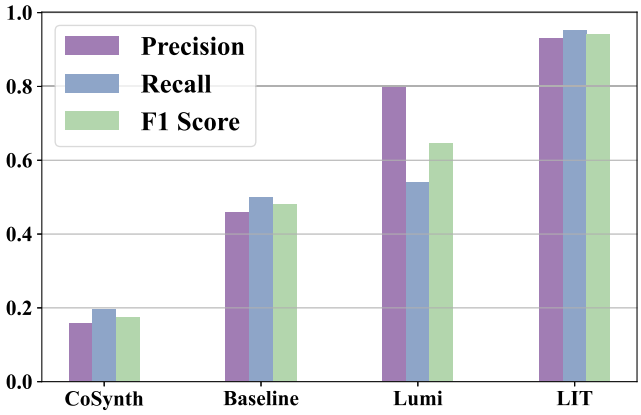


Fig. 8. The comparison of the Precision, Recall, and F1 Score results for intent translation between CoSynth, ChatGLM-6B (baseline), Lumi, and LIT on the Lumi$^+$ dataset.

The network state $\mathcal{S}_v$ when a user service arrives consists of two main components: (1) the user service requirement, represented by a two-element vector $\{s_v, b_v\}$, the source node and the destination node of service v. (2) the network resource state, which describes the current network status and includes the available bandwidth capacity for each network link $\{D_l : l \in L\}$. The action space, $\mathcal{A} = D_a = \{d_1, d_2, \ldots, d_a\}$ is defined by the bandwidths that can be set by the network links. The rewards, represented as $r_v = R_{v,f_k}$, refer to the benefits of each QoS indicator derived from the interaction between the model and the environment. These rewards must also adhere to the constraint specified in (4d).

In order to facilitate comprehensive examination of the network state and edge space while enhancing model learning stability, a noise layer has been integrated into the DQN, replacing the conventional ϵ-greedy strategy [54]. The parameters of both the behavior and target networks are augmented with stochastic variables ϵ and ϵ^-, in conjunction with noise parameters σ and σ^-. Consequently, the objective function of the DQN is formulated as follows:

$$L(\theta) = E_{\epsilon^-,\epsilon}\Big[E_{(\mathcal{S}_v,a_v,r_v,\mathcal{S}_{v+1})\sim D}\Big[r_v + \gamma \max_{a^*\in\mathcal{A}} Q\left(\mathcal{S}_{v+1}, a^*, \epsilon^-; \theta^-, \sigma^-\right) - Q\left(\mathcal{S}_v, a_v, \epsilon; \theta, \sigma\right)\Big]^2\Big] \tag{11}$$

Where θ refers to the model parameters of the behavior network, while θ^- indicates the model parameters of the target network. The symbol $\mathcal{S}_{v+1}$ denotes the next state of the network. The variable a_v represents an action that can be executed given the current network status, whereas a^* denotes an action that is available in the subsequent new network status. The parameter γ represents the discount factor.

These models collectively constitute a set of expert models. By leveraging the analytical capabilities of LLM, the required expert models can be directly deduced from user requirements, facilitating the assignment of a corresponding weight matrix $W = \{w_1, w_2, \ldots, w_k\}$ to the selected expert models, where k denotes the k-th QoS indicators. Each expert model generates its own Q value vector in response to the provided input network state, represented as $Q(\mathcal{S}_v, a_v) = \{q_1, q_2, \ldots, q_a\}$. The resulting

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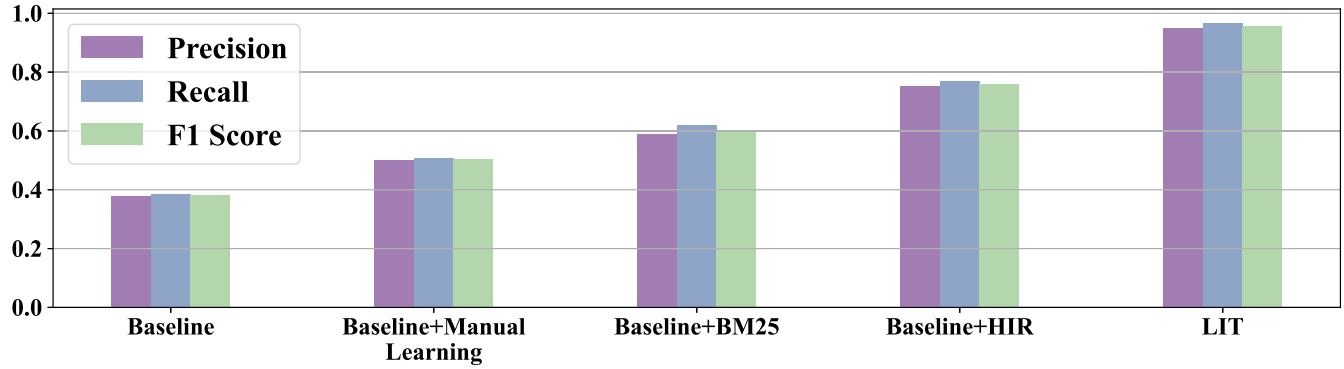


Fig. 9. The comparison of the Precision, Recall, and F1 score of the intent translation task with the introduction of manual learning and RAG.

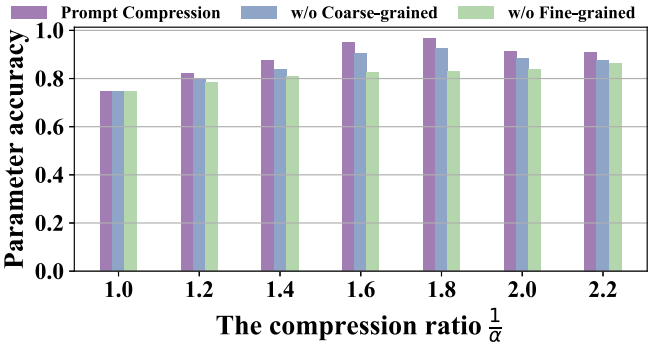


Fig. 10. The parameter accuracy of prompt compression under different compression ratio $\frac{1}{\alpha}$.

TABLE III
THE DETAILS OF DATASETS

Dataset	Component	Number
LIT	Mappings from natural language intent to PS	1600
	Explanation of policies and parameters	500
	Relationships between policies and parameters	450
Lumi$^+$	Original Lumi	200
	Expanded Lumi	800

weighted Q value vector is given by:

$$\hat{Q}(\mathcal{S}_v, a_v) = W \cdot \begin{bmatrix} Q_1(\mathcal{S}_v, a_v) \\ Q_2(\mathcal{S}_v, a_v) \\ \vdots \\ Q_k(\mathcal{S}_v, a_v) \end{bmatrix} \tag{12}$$

The final selected action $\hat{a}$ is described by the following equation:

$$\hat{a} = \max_{a_v} \hat{Q}(\mathcal{S}_v, a_v) \tag{13}$$

the variable $\hat{a}$ represents the parameter value at the metric level specifically for bandwidth. This approach can be similarly extended to other metric-level PVG tasks.

V. EVALUATION

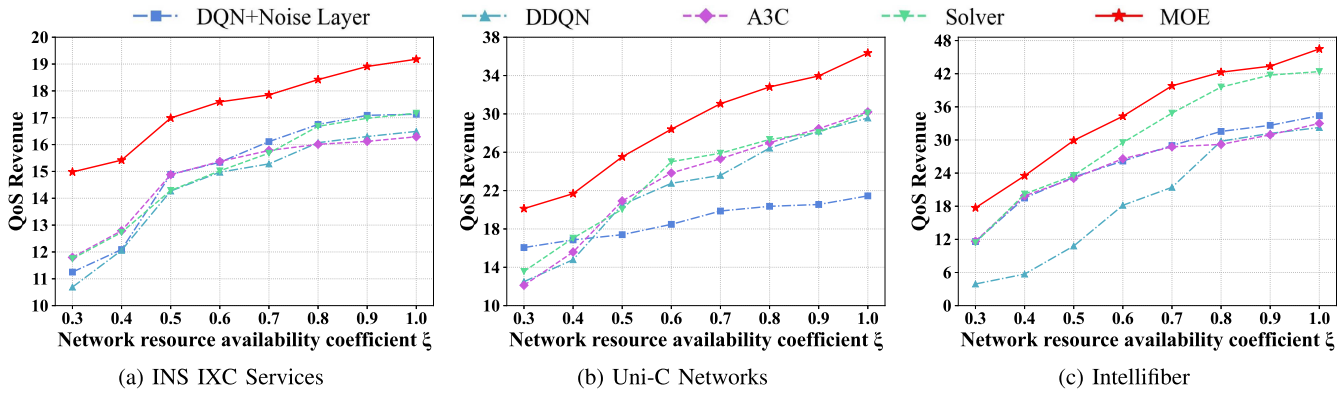
A. Experiment Setup

Simulated network setting: We conduct relevant experiments using real-world network topologies of three different scales: INS IXC Services, Uni-C Networks, and Intellifiber [55]. The INS IXC Services topology comprises 33 nodes and 41 links, Uni-C Networks consists of 51 nodes and 64 links, while Intellifiber includes 72 nodes and 85 links. In the INS IXC Services topology, we configured 10 user nodes. Similarly, in the Uni-C Networks topology, the number of user nodes is set to 20. For Intellifiber, we established 30 user nodes. The initial bandwidth, signal power, and noise power for each link are set at 100 Mbps, 1000 W, and 1 W, respectively. The available bandwidth options for each link are $[5\ M, 10\ M, 15\ M, 20M]$. We introduce a new parameter, ξ, which denotes the network resource availability coefficient, with values ranging from 0.3 to 1.0. For instance, when $\xi = 0.5$, the links have a maximum capacity of 50 Mbps, whereas when $\xi = 1.0$, the maximum capacity of the links increases to 100 Mbps. All experiments are conducted on a machine equipped with an Intel(R) Xeon(R) Silver 4210R processor, 503 GB of memory, and an NVIDIA A100 GPU (80 GB VRAM).

Relevant parameter setting: To facilitate intent translation, we have assembled a collection of open-source wireless network management commands and manuals to serve as LIT datasets and a knowledge base. This collection contains 1,600 mappings from natural language intent to PS, with 90% reserved for PSG training. Furthermore, there are 500 sets describing policies and parameters, and 450 sets outlining the connections between policies and parameters, all of which constitute the dataset for Manual Learning. Additionally, we have expanded the Lumi dataset from 200 to 1000 entries, resulting in the Lumi$^+$ dataset, which is used to assess LIT's performance on the Lumi dataset, utilizing 90% of the data for model training. The expansion of the Lumi dataset is carried out through manually curated data. Details of the dataset composition are presented in Table III.

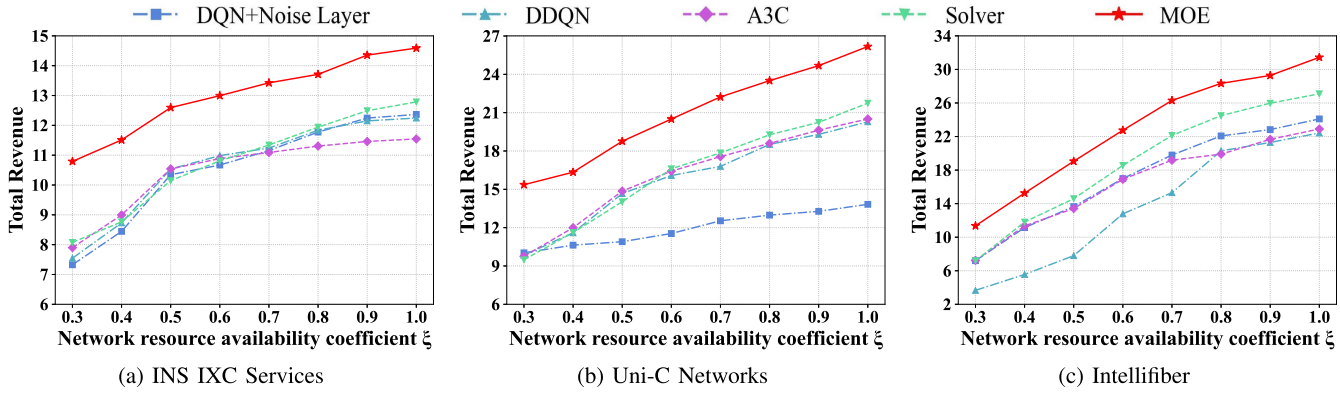
Within the LIT datasets, the model's task is to generate a comprehensive policy set based on the provided intent. The Lumi$^+$ dataset is associated with the PVG task, where the model is responsible for filling in parameters according to the intent.

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(a) INS IXC Services (b) Uni-C Networks (c) Intellifiber

Fig. 11. The results of QoS revenue under different scale network topologies.



(a) INS IXC Services (b) Uni-C Networks (c) Intellifiber

Fig. 12. The results of total revenue under different scale network topologies.

For PSG, we employ the ChatGLM-6B [17] model for full-parameter fine-tuning, using the Adam optimizer with a learning rate of 10^{-4}. In PVG, ChatGLM-6B is also fine-tuned after hierarchical manual indexing to produce PVs. In the MoE, parameters are set as follows: $\eta_1 = 1$, $\eta_2 = 0.25$, $\lambda_1 = \lambda_2 = 0.5$, and $\tau = 10$. The DQN models are trained with the Adam optimizer at a learning rate of 10^{-3}, $\gamma = 0.1$, a batch size of 64, over 300 epochs, with the update interval between the estimated Q network and target Q network parameters set to 1000. Selection of user and server nodes is based on the user's intent, followed by calculation of the shortest path to determine the base stations to be traversed. We train a set of expert models using DRL or MILP solver, aiming to maximize the advantages of each QoS indicator by choosing the optimal bandwidth. As the optimization goal for the expert model, key QoS indicators considered in the network include Data Transfer Rate (DTR), Packet Loss Rate (PLR), and Bit Error Rate (BER), which are primary indicators of QoS in data transmission. To assess the impact of prompt compression, we evaluated the related documents for 200 parameter queries and verified the accuracy of the final values for these parameters. ChatGLM-6B is utilized as the model for perplexity prediction. Parameters for prompt compression are configured as follows: top $M' = 5$ and $\delta_\alpha = 0.3$.

Evaluation metrics: The accuracy of intent translation by LLM is evaluated by comparing the final commands and PVs it generates. This assessment is based on three metrics: Recall, Precision, and F1 score. Recall is determined by $\frac{N_r}{N}$, where N represents the total number of parameters in the test set and N_r denotes the parameters that perfectly align with the ground truth. Precision is computed as $\frac{N_r}{N_p}$, with N_p being the total number of parameters generated by LLM. The F1 score is derived using the formula $\frac{2 \cdot R \cdot P}{(R+P)}$. The metric-level PVs take into account the total QoS revenue, which combines the benefits of three QoS indicators as outlined in (2). They also consider the total revenue, which includes SSR rewards and QoS benefits as defined in (4a). These serve as the primary metrics for assessing the effectiveness of the MoE approach. For prompt compression, we assess its efficacy by measuring parameter accuracy, which reflects the count of correct values among the parameters evaluated.

B. Performance of LIT

To demonstrate the end-to-end performance of LIT on the intent translation task, we selected the LIT datasets and the Lumi$^+$ dataset for experiments. We trained ChatGLM using full-parameter fine-tuning on the task, utilizing PSG and PVG for the training process. The fine-tuned model was then established as the baseline. For Baichuan [56], we applied LoRA [57] training based on PSG and PVG. As for CoSynth [27], only its intent translation component was trained. Lumi was trained

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TABLE IV
THE EXAMPLE OF COMPARING OUTPUTS FROM DIFFERENT SOLUTIONS

User Intent: Add a wireless base station to a university campus, primarily for web browsing and video conferencing services.
LIT: ADD SECTOR: SectorId=4, ANTNUM=2, createSectorEqm=TRUE, SectorEqmId=16; ADD CELL: LocalCellId=1, CellName='university campus', SectorId=4, FreqBand=30, DlEarfcn=44308, UlBandWidth=CELL_BW_N100, DlBandWidth=CELL_BW_N100, FddTddInd=CELL_TDD, HighSpeedFlag=HIGH_SPEED, UePowerMaxCfgInd=NOT_CFG; ADD EUCELLSECTOREQM: LocalCellId=1, SectorEqmId=16; ACT CELL: LocalCellId=1;
Baseline+HIR: ADD CELL: LocalCellId=1, CellName='university campus', SectorId=4, FreqBand=30, DlEarfcn=44308, UlBandWidth=CELL_BW_N100, DlBandWidth=CELL_BW_N100, FddTddInd=CELL_TDD, HighSpeedFlag=HIGH_SPEED; **Three policies are missing.**
Baseline+Manual learning: ADD SECTOR: SectorId=4, ANTNUM=2, createSectorEqm=TRUE, SectorEqmId=1; ADD CELL: LocalCellId=1, CellName='university campus', SectorId=0, FreqBand=30, DlEarfcn=1, UlBandWidth=100M, DlBandWidth=100M, FddTddInd=CELL_FDD, HighSpeedFlag=LOW_SPEED; ADD EUCELLSECTOREQM: LocalCellId=1, SectorEqmId=1; ACT CELL:LocalCellId=1;
Baseline: ADD CELL: LocalCellId=1, CellName='university campus', SectorId=0, FreqBand=30, DlEarfcn=1, UlBandWidth=100M, DlBandWidth=100M, FddTddInd=CELL_FDD, HighSpeedFlag=LOW_SPEED; **Three policies are missing.**
CoSynth: CREATE ENB: ENBId=0, ENBName='university campus', BandWidth=50M, FREQBANDIND=1; ACTIVE ENB: ENBId=0;

in the same way as described in the original paper. Precision, recall, and F1-score are chosen as evaluation metrics for the comparative experiments. Based on different datasets, the experimental results are shown in Figs. 7 and 8. On the LIT dataset, LIT improved the baseline in Precision, Recall, and F1 score by 56.98%, 58.03%, and 57.5% respectively. On the Lumi^{+} dataset, LIT improved the baseline in Precision, Recall, and F1 score by 47.02%, 45.29%, and 46.0% respectively. A comprehensive comparison of the experimental results demonstrates that LIT exhibits robust intent translation capabilities, with notable enhancements in comparison to alternative models. Due to its lack of comprehensive parameter fine-tuning support and reliance on manual learning, CoSynth exhibited comparatively weaker performance. Despite Baichuan reaching convergence in its loss function, it still underperformed in comparison to ChatGLM, primarily due to the restricted number of parameters that could be adjusted using LoRA. The Baseline model exhibited deficiencies in the incorporation of manual knowledge and the analysis of service intent, which resulted in the emergence of grammatical errors and invalid PVs. Given that Lumi employs an LSTM architecture and lacks the capacity for policy generation, experiments were not conducted on the LIT dataset. This also underscores the limited transferability of Lumi.

C. Performance of RAG

We design ablation experiments to demonstrate the effectiveness of manual learning in PSG and RAG in PVG, with results shown in Fig. 9. After incorporating manual learning, the F1 improved by 32.08% compared to the baseline. We optimized the overall framework of RAG and incorporated experimental comparisons of BM25 [58], HIR algorithms, and LIT within the RAG section of the ablation experiments. Additionally, to prove the positive impact of the prompt compression algorithm on generating semantic-level PVs by LLMs, we compared the PVs generation effects at different compression ratios $\frac{1}{\alpha}$, as illustrated in the Fig. 10.

An example of comparing outputs from different solutions is shown in Table IV, where discrepancies in the outputs are highlighted in red font. LIT's output matches the true labels exactly. The output from Baseline+HIR lacks three policies, 'ADD SECTOR', 'ADD EUCELLSECTOREQM' and 'ACT CELL'. The output from Baseline+Manual learning contains some PVs errors, such as 'SectorEqmId=1' and others. The baseline can generate policies that generally comply with the manual's requirements, but it lacks some policies and parameters. Due to having undergone only instruction, CoSynth produces policies of lower quality.

Based on the analysis of the above experimental results, we can draw the following conclusions:

- The performance of LIT is superior to that of other end-to-end methods in the domain of intent translation. During the process of PSG, LIT guarantees that the PS is syntactically correct and logical, thereby establishing the foundation for overall correctness. LIT generates the PVs with greater precision, which ultimately results in the production of a high-quality policy set.
- The absence of manual learning impedes the model's capacity to generate new policies during the PSG phase. This indicates that manual learning can augment PSG performance by enabling the model to glean insights from the corpus of manuals, thereby mitigating error propagation.
- The absence of RAG frequently results in the generation of invalid PVs during the PVG phase, which in turn causes policies to be undeployable. This phenomenon can be attributed to the illusions problem of LLMs. Furthermore, the lack of manual learning also affects PVG phase.
- The choice of retrieval algorithm can affect the final generation results of semantic-level PVs. Consequently, the RAG process must be optimized in accordance with the characteristics of the manuals in question, for example, by implementing HIR. Furthermore, the compression of retrieval results will result in a reduction of noise and an enhancement of LLMs' performance in generating PVs.

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D. Performance of MoE

The introduction of MoE has the effect of significantly alleviating the mismatch between user demands and network conditions, thereby reducing the likelihood of resource conflicts. We compared MoE with four optimization models (including three different DRL models and a MILP solver) in terms of their impact on overall QoS and total revenue across different network topology scales, as shown in Figs. 11 and 12. The input and output of the DQN, DDQN, and A3C models are the same as those of the expert model. Results analysis indicates that, with a higher network resource availability coefficients (ξ=0.3), the MoE method increases QoS revenue by 26.95%, 25.28% and 51.75% in the INS IXC network, Uni-C network, and Intellifiber, respectively, as shown in Fig. 11. The experimental results demonstrate that by understanding user intent, the LLM is capable of accurately adjusting the weights of the outputs of various expert models across different topology scales.

To enable comparison from multiple perspectives, we used total revenue as an evaluation metric and conducted additional experiments on the aforementioned models. The experimental results are shown in Fig. 12. When ξ reaches 0.3, the MoE method increases total revenue by 33.62%, 53.14%, and 56.95% in the INS IXC network, Uni-C network, and Intellifiber, respectively, compared to the second-best method.

VI. Conclusion

This paper proposes LIT, an LLM-based framework for intent translation. Large language models demonstrate strong performance in intent translation tasks due to their exceptional understanding and generation capabilities. However, IBN still faces two major challenges: hardware heterogeneity and highly dynamic networks. LIT introduces RAG to enhance the LLM's ability to generate semantic-level PVs by retrieving information from heterogeneous hardware manuals. Considering the structured nature of these manuals, we propose a HIR algorithm for efficient and accurate retrieval, further compressing the retrieval results to reduce noise. To align the generated configurations with dynamic network states, LIT incorporates MoE to produce metric-level parameters. This approach leverages the LLM to generate a weight matrix based on service requirements and the optimization directions of expert models, synthesizing the outputs of multiple expert models to produce metric-level parameters. Experimental results show that LIT improves the F1 score by 57.5% compared to the baseline. Analyzing the results demonstrates that LIT effectively mitigates the challenges faced by IBN, generating strategies that adapt to downstream networks and hardware by deeply understanding user intent.

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